TrustSense-BMS: A Physics-Informed AI and Blockchain Integrated Battery Management System

1. Abstract

Modern Electric Vehicle (EV) batteries face significant challenges regarding thermal safety, network cybersecurity, and data provenance. Current Battery Management Systems (BMS) rely heavily on expensive physical temperature sensors, lack robust intrusion detection for CAN bus networks, and use mutable databases that cannot guarantee compliance with upcoming EU Battery Passport regulations. This project proposes TrustSense-BMS, an integrated platform that addresses these gaps using a four-pillar architecture. It utilizes a Physics-Informed Graph Neural Network (GNN) to act as a "Virtual Sensor," reducing physical sensor dependency by mathematically estimating cell temperatures. A Time-Series Transformer monitors CAN traffic to detect cyberattacks such as spoofing or flooding. Critical telemetry and incident hashes are logged onto a permissioned Hyperledger Fabric blockchain to ensure an immutable audit trail. Finally, a Retrieval-Augmented Generation (RAG) LLM assistant provides secure, plain-language engineering queries based entirely on verified blockchain data and technical documentation.

2. Introduction

The rapid adoption of Electric Vehicles necessitates advanced battery management to prevent thermal runaway. However, adding physical sensors to every cell increases manufacturing costs, weight, and potential points of failure. Simultaneously, as vehicles become highly connected, the internal CAN/CAN-FD bus network becomes vulnerable to cyber-physical attacks. Furthermore, regulatory bodies, such as the EU, mandate a permanent, verifiable history of the battery (Digital Battery Passport) for recycling and secondary market valuation, effective February 2027. TrustSense-BMS acts as a "Trust Layer" and "Physics Layer" overlay for traditional BMS to address these critical pain points.

3. Problem Statement vs. Proposed Solution

3.1 Existing System Challenges

Challenge Area	Description
Sensor Dependency	Current systems rely on high physical sensor counts; sensor failure creates dangerous thermal blind spots.
Lack of Cyber Security	Standard BMS controllers trust CAN bus data implicitly and cannot distinguish between hardware faults and malicious spoofing.
Mutable Data Records	Battery histories stored in centralized SQL databases are susceptible to tampering, invalidating data for warranty and regulatory audits.

3.2 Proposed System Innovation

TrustSense-BMS shifts intelligence from hardware to software through four technical pillars:

Objective	Metric/Goal
Reduce physical temperature sensor dependency	>50% reduction with RMSE ≤ 1.0°C
Detect CAN network intrusion	Detection delay ≤ 500ms; False positive rate ≤ 5%
Data Provenance	Cryptographically secure log compliant with EU Battery Passport
Telemetry Analysis	Democratized access via RAG-powered engineering assistant

1. **AI Virtual Sensor:** Predicts temperatures for unsensed cells using Physics-Informed GNNs/PINNs to reduce hardware costs.
2. **Cyber IDS:** Uses Time-Series Transformers and Isolation Forests to learn normal rhythm and physics boundaries, flagging thermodynamically impossible data jumps as attacks.
3. **Blockchain Provenance:** Commits SHA-256 hashes of telemetry batches and incident logs to Hyperledger Fabric for an un-hackable audit trail.
4. **Fullstack RAG Assistant:** A localized Llama 3 assistant allows operators to query blockchain logs and manuals via a React-based dashboard.

4. Project Objectives

1. To reduce physical temperature sensor dependency by at least 50% while maintaining a Root Mean Square Error (RMSE) of ≤ 1.0°C.
2. To detect CAN network intrusion and data spoofing with a detection delay of ≤ 500ms and a false positive rate of ≤ 5%.
3. To provide a cryptographically secure, immutable audit log compliant with future Battery Passport regulations.
4. To democratize complex telemetry analysis via a natural language engineering assistant.

Component	Technology / Tooling
Data Generation	PyBaMM (Synthetic Thermal Fields)
Artificial Intelligence	PyTorch Geometric (GNN), Transformers, Scikit-learn
Blockchain Ledger	Hyperledger Fabric, Node.js (Chaincode)
Backend & LLM	FastAPI, LangChain, ChromaDB, Local Llama 3
Frontend UI	React.js (3D Heatmaps & Threat Tickers)
Deployment	Ubuntu 22.04 LTS, NVIDIA Jetson Orin Nano (Edge AI)

5. Technology Stack & Methodology6. Safety & Operational Constraints

Advisory Overlay: The system is strictly a monitoring-only advisory platform. It provides dashboards, alerts, and immutable logs but **shall NOT** issue hardware-level commands or control high-voltage contactors, ensuring isolation from the real-time safety control loop.